

BIO262
PHYSIOLOGY

MODULE 6 -

BRAIN

ACTION POTENTIALS

**AUTONOMIC NERVOUS
SYSTEM**

SMELL AND TASTE

Fancy way to explain how a neuron sends a message – this is an electrical current/impulse which opens **voltage** sensitive gates in the axolemma allowing Na^+ and K^+ to move in and out of the axon.

When axon is **myelinated** the impulse moves very rapidly!

Action potentials jump down axon.

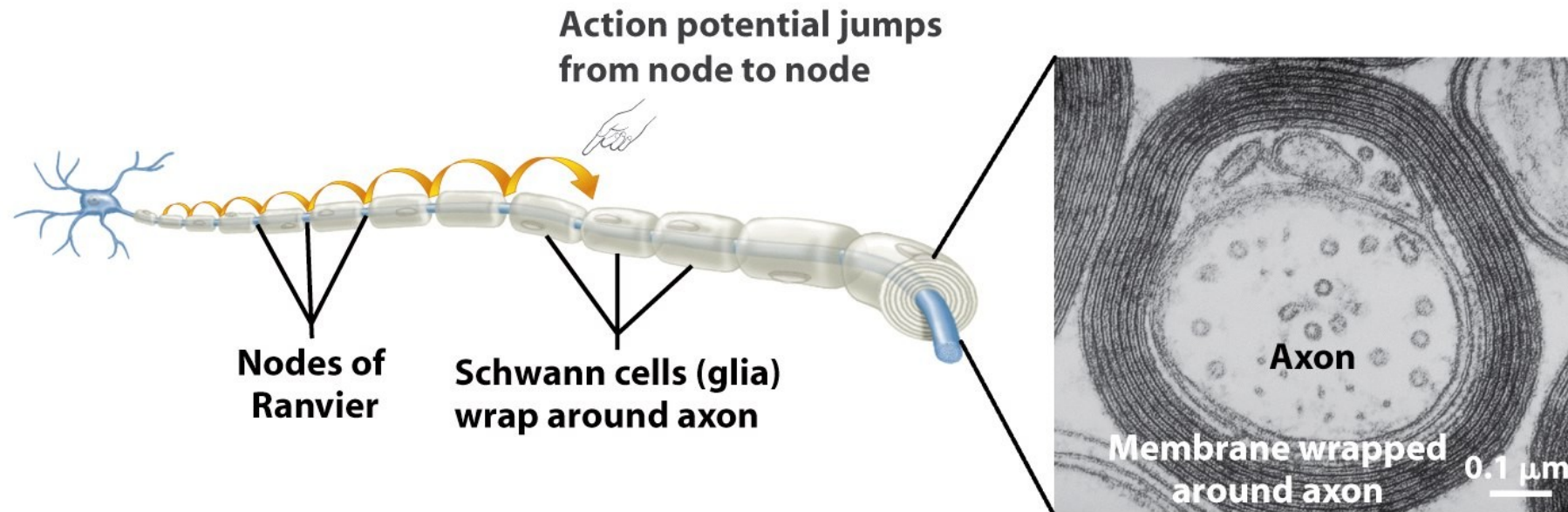
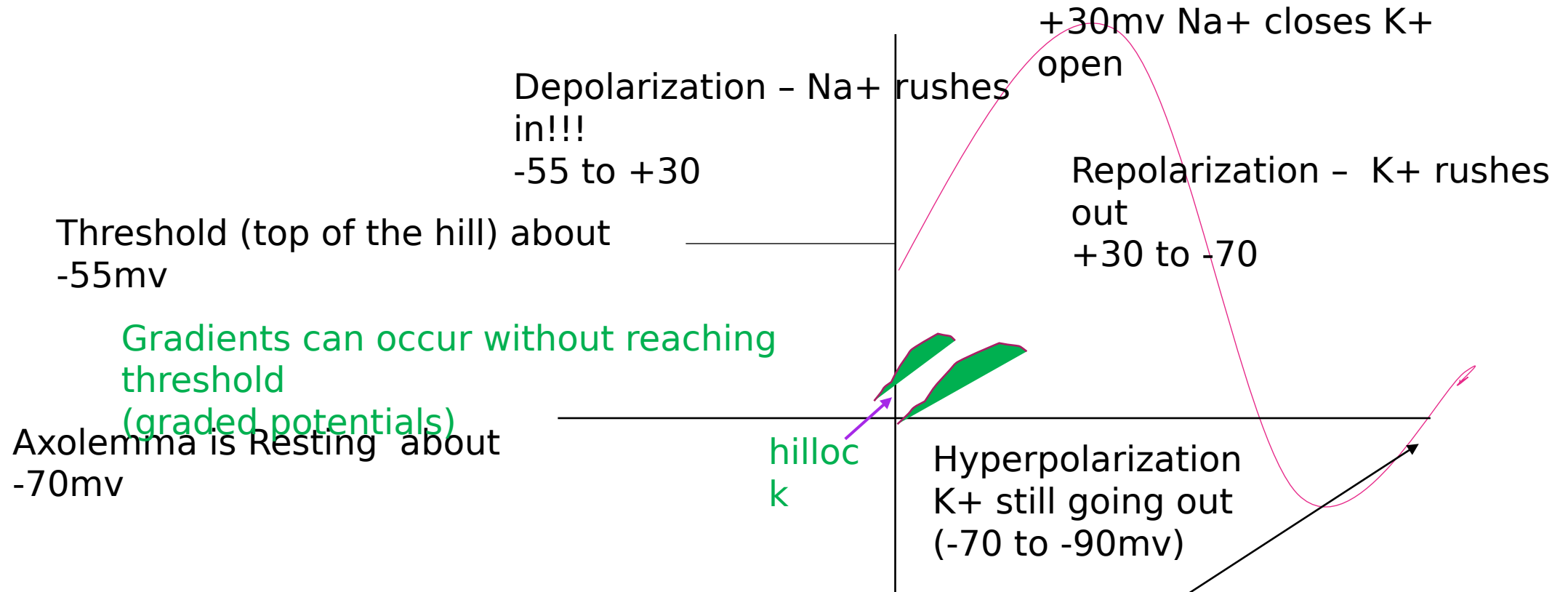


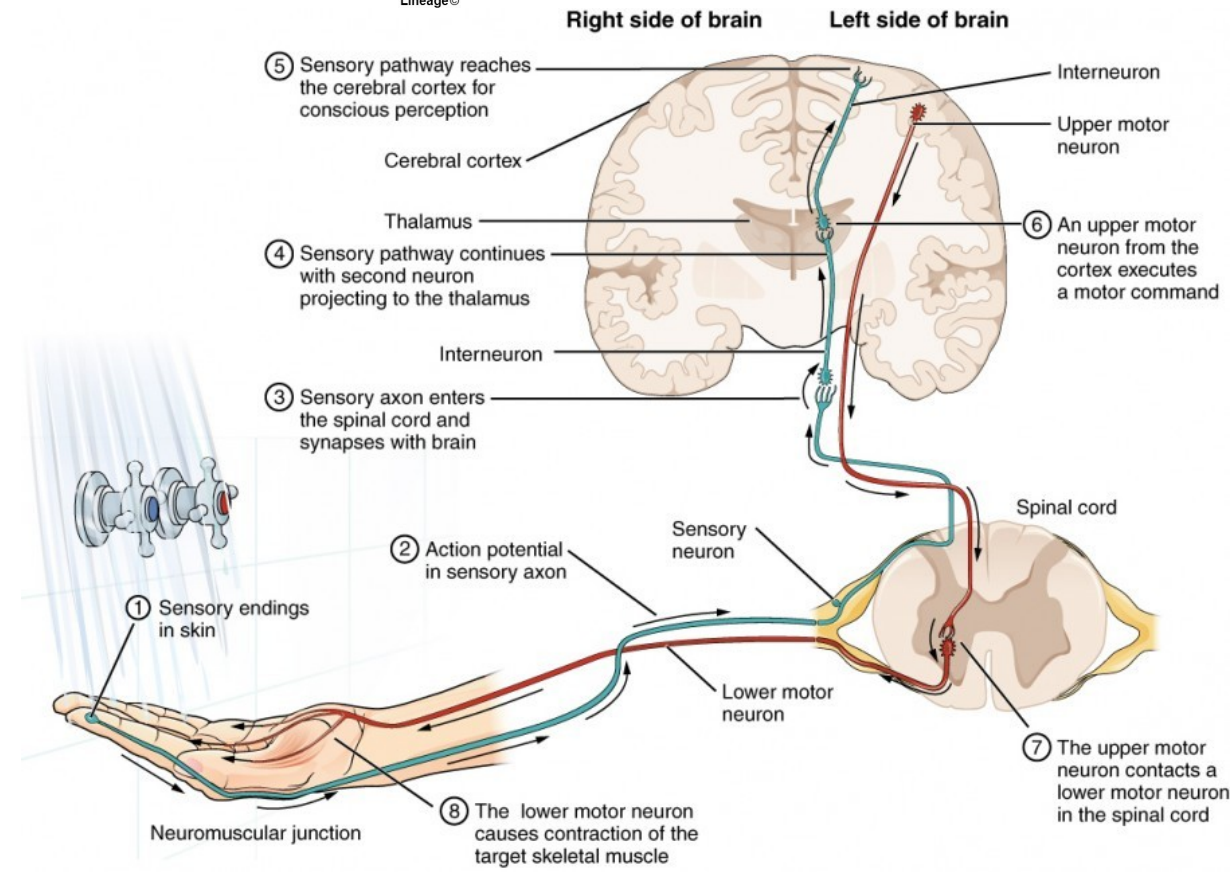
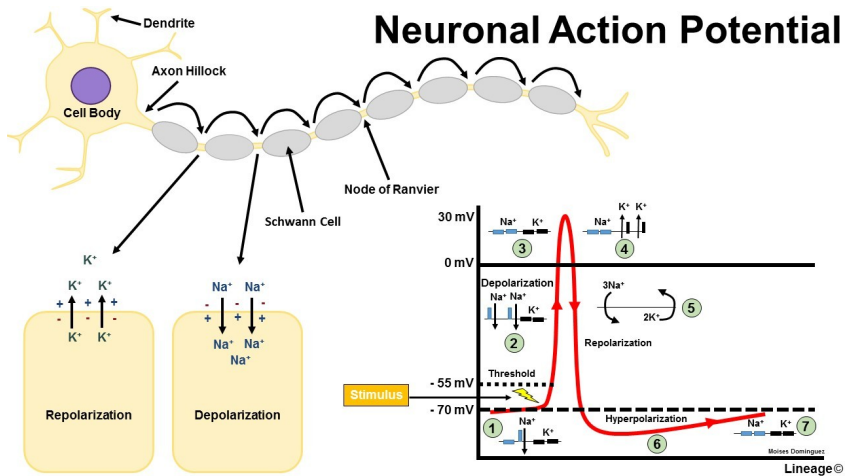
Figure 45-12a Biological Science, 2/e
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GRADIENTS; GRADED AND ACTION POTENTIALS



To get the membrane back to RESTING Na⁺/K⁺ pump furiously works to pump Na⁺ OUT and K⁺ IN.

This nerve CANNOT fire until returns to normal resting.



Sensation

Sensory receptor

Sensory neuron

CNS

-spinal cord to brain,
brain processes, then
sends motor through
spinal cord to:

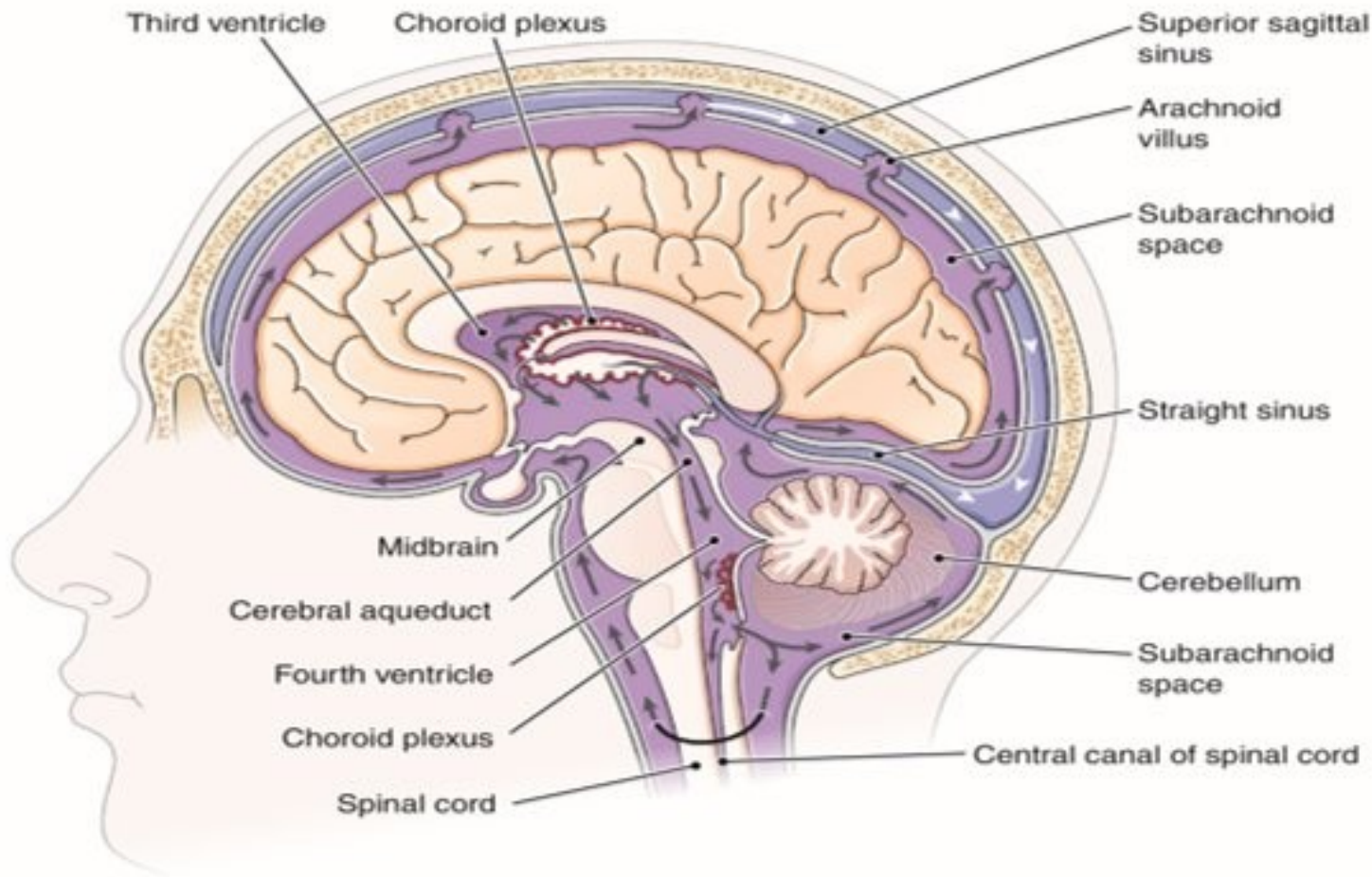
Motor neuron

Effector (muscle or
gland)

Response

LET'S LOOK AT THE BRAIN

CHOROID PLEXUS AND VENTRICLES



What makes CSF?

How does it
drain?/where?

What's its job?

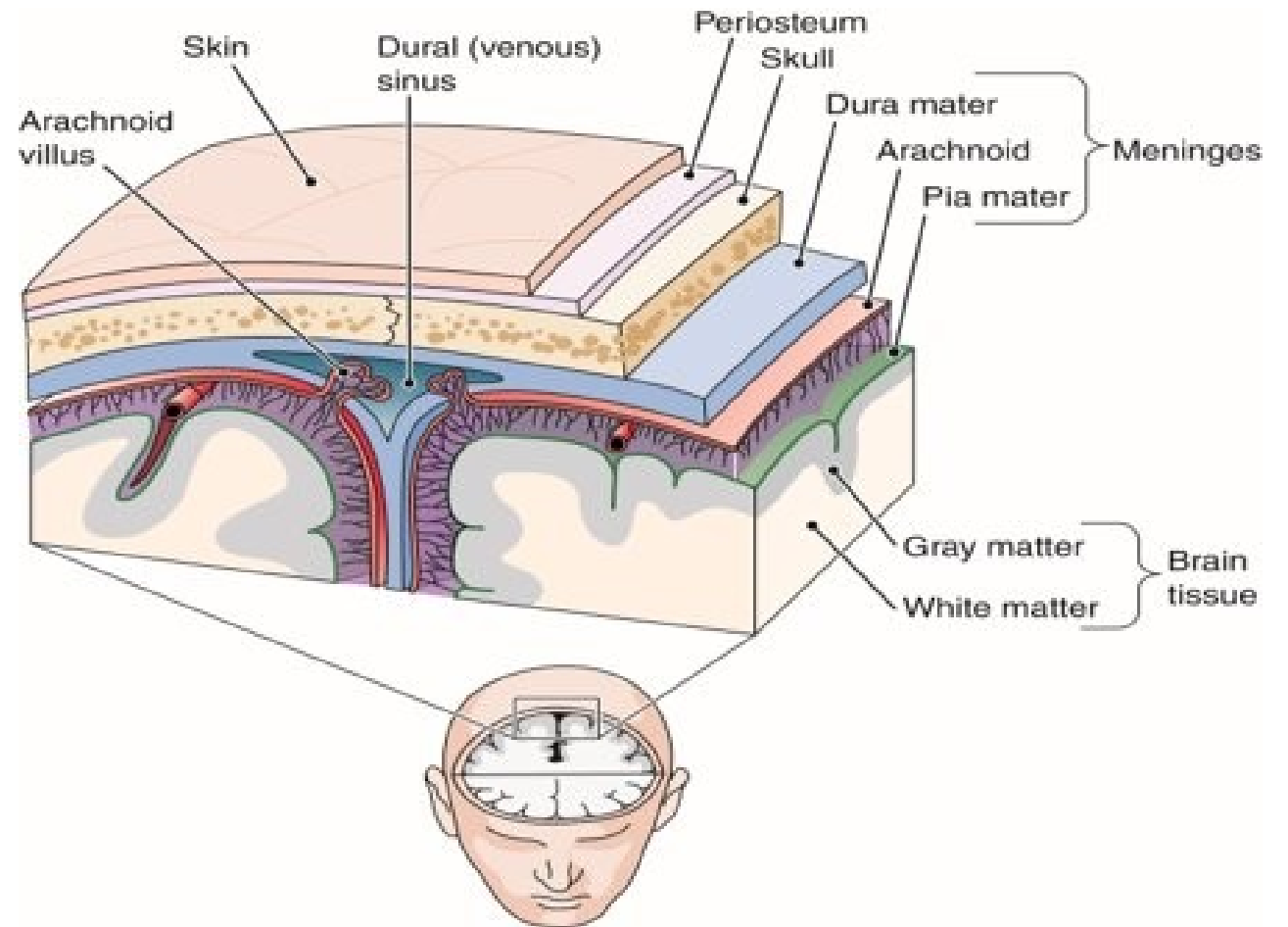


MENINGES

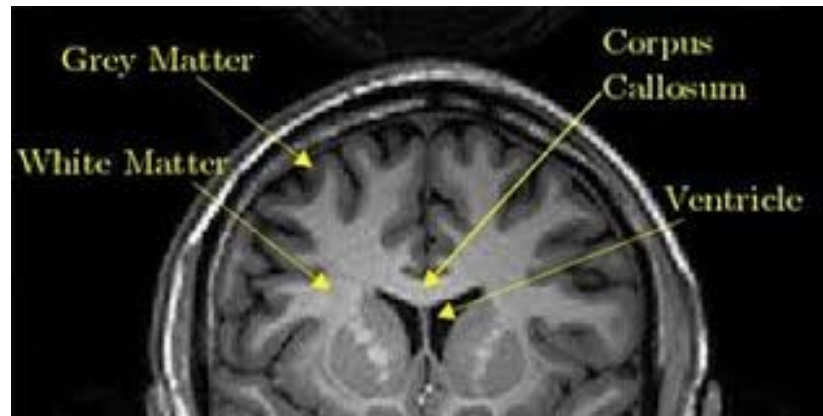
WHAT'S THE RIGHT ORDER?

- Arachnoid Mater
- Dura Mater
- Subarachnoid Space
- Pia Mater

- **JOB?**



WHITE VERSUS GREY MATTER



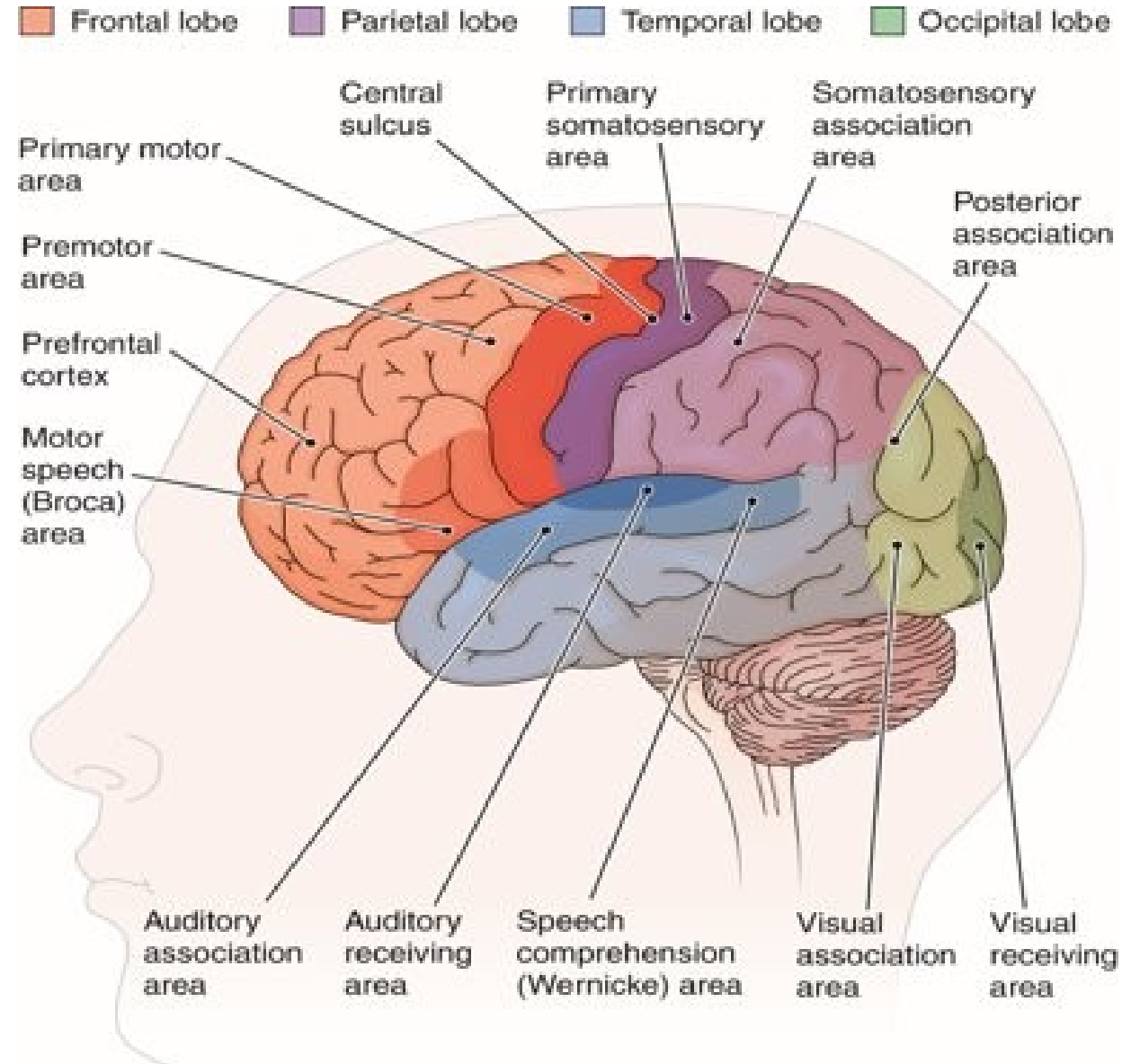
White matter=myelinated axons
(sends information/communication
between brain areas)

Grey matter = cell bodies where
information is processed.

BRAIN LOBES

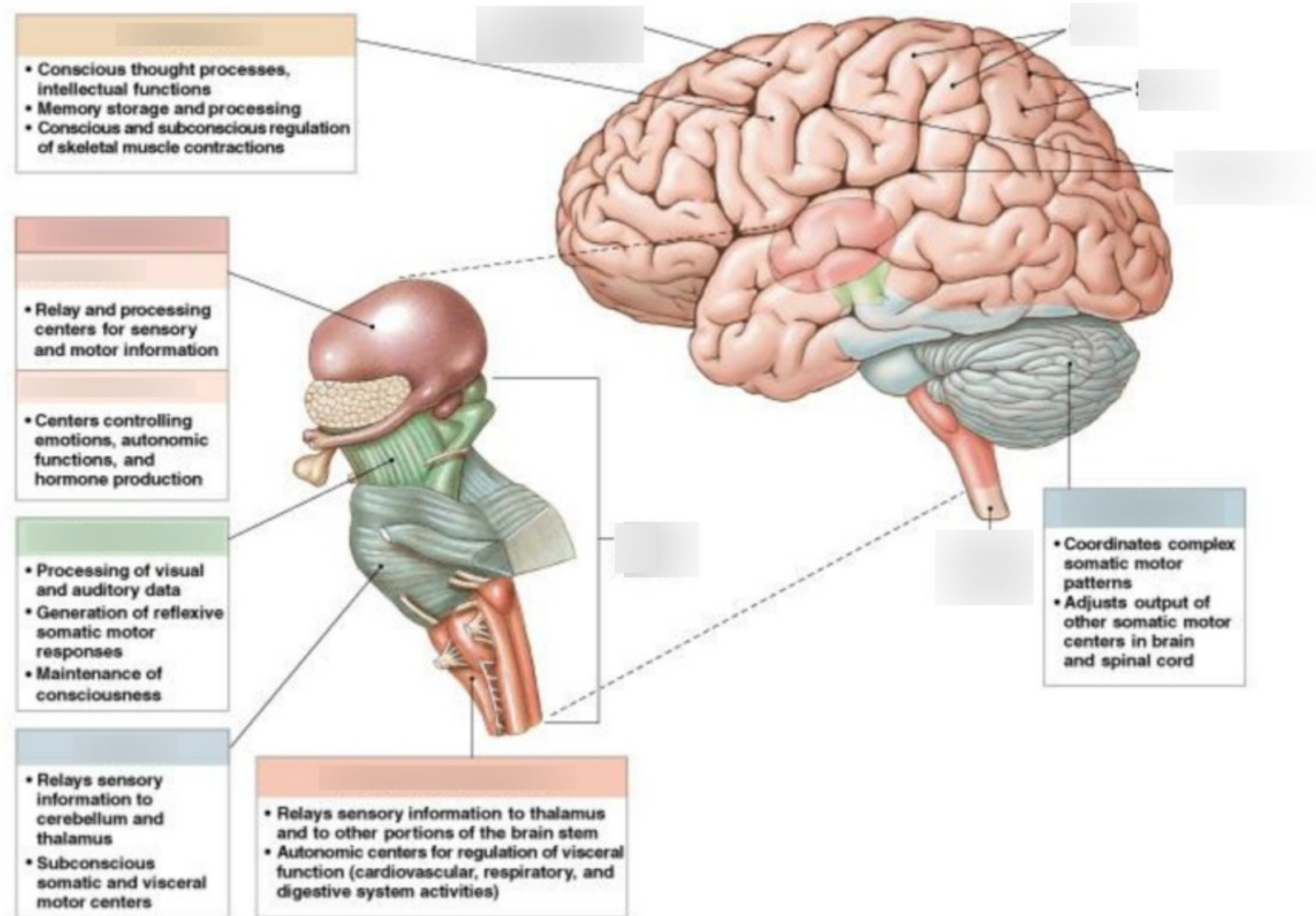
WHAT DOES WHAT?

- Vision
- Smell
- Coordinates movement/**balance**
- Taste
- **Decision making**/consequences
- **Hearing**
- Speech: motor vs understanding



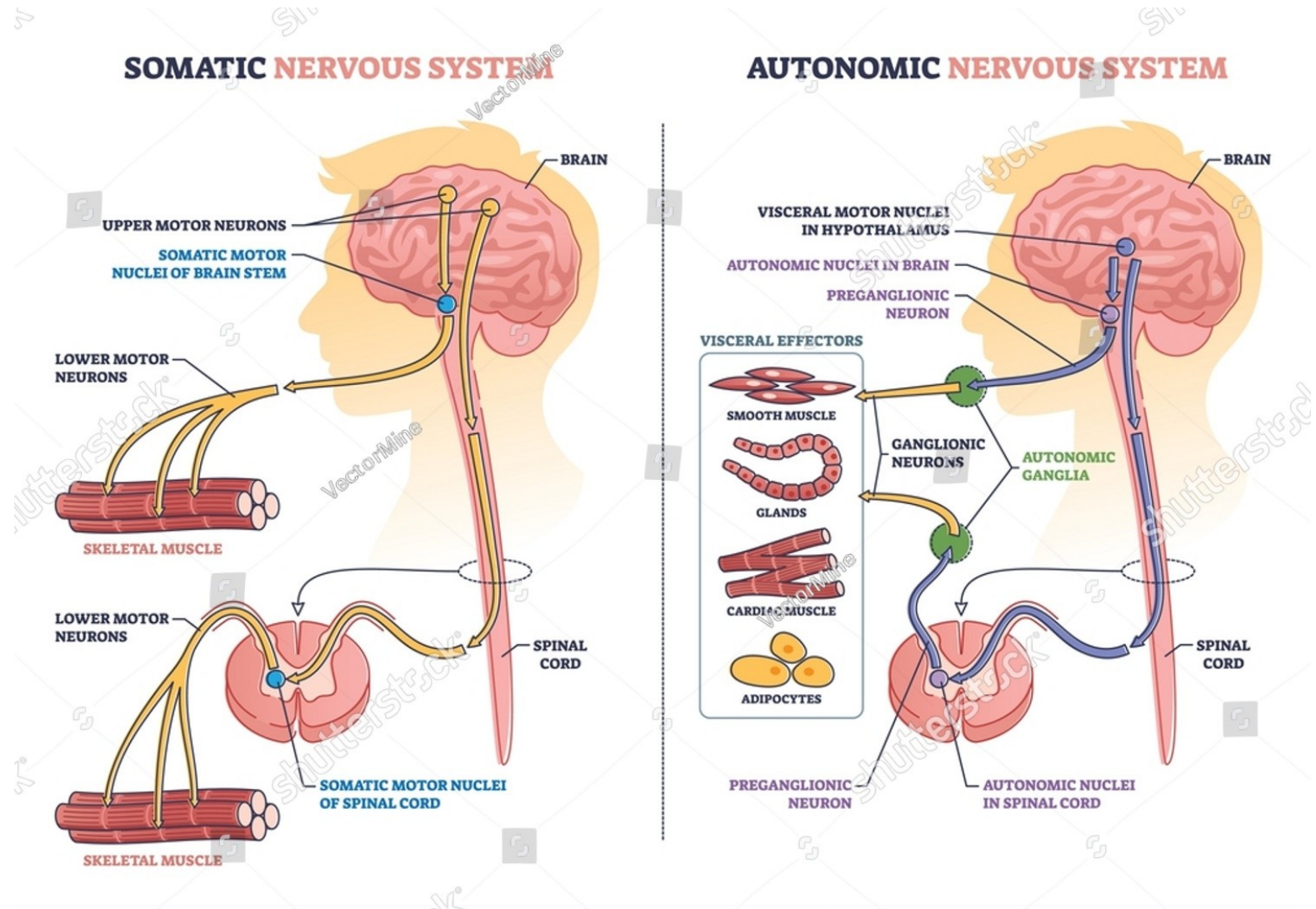
Brain Stem:
Controls vital
life functions
such as:
-breathing
-heart rate

Thalamus:
relay/processor
center for all
sensory
information



Somatic =
voluntary
skeletal
muscles

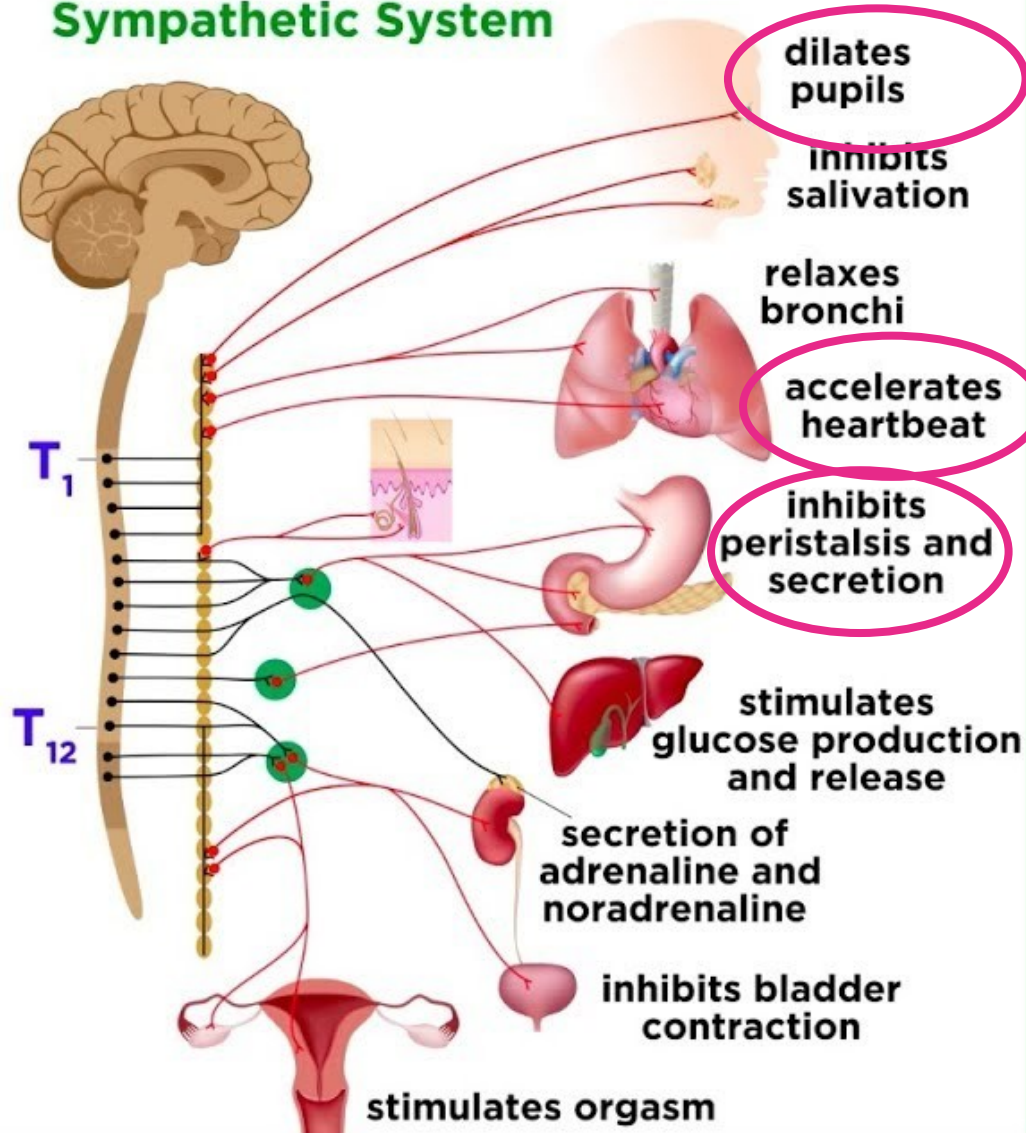
Autonomic =
Involuntary
Smooth
muscles
glands



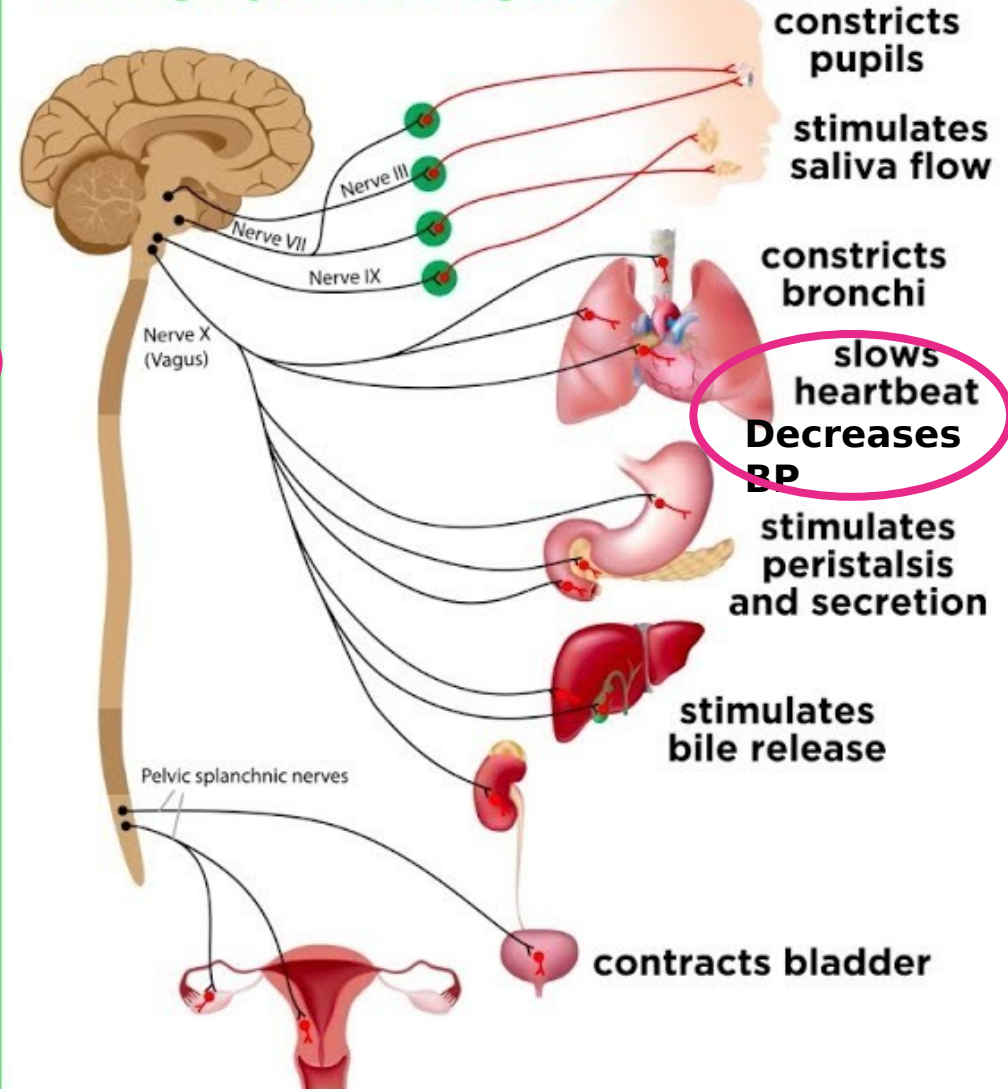
Fight or flight

Rest and digest

Sympathetic System



Parasympathetic System



OLFACTORY (SMELL) ORGANS

Olfactory receptors—

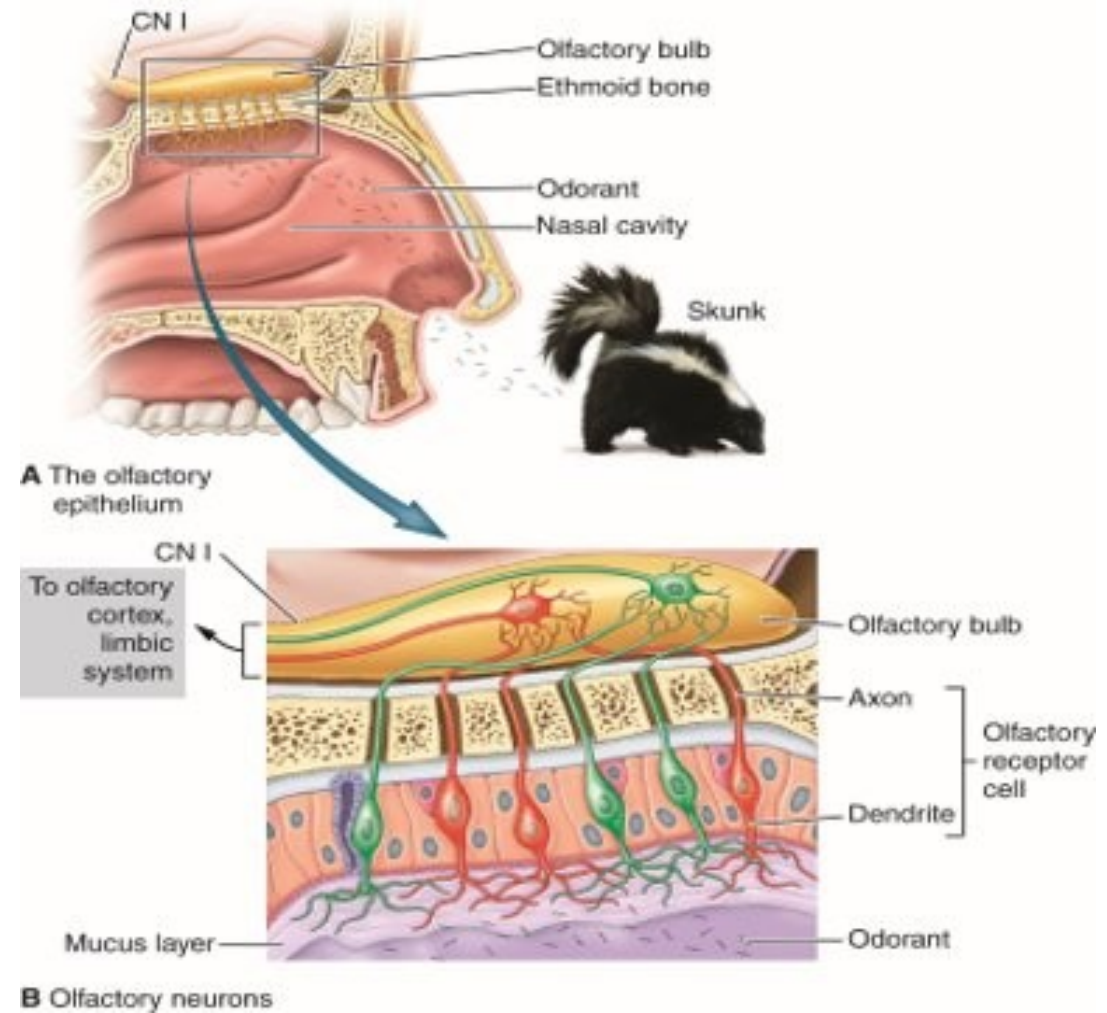
dendrites of neurons found in the olfactory epithelium.

(CHEMORECEPTOR – Phasic)

Olfactory glands—found in

lamina propria; secretions absorb water and form a thick colored mucous

<https://www.youtube.com/watch?v=0xclbPkzN9w>



OLFACTORY PATHWAY

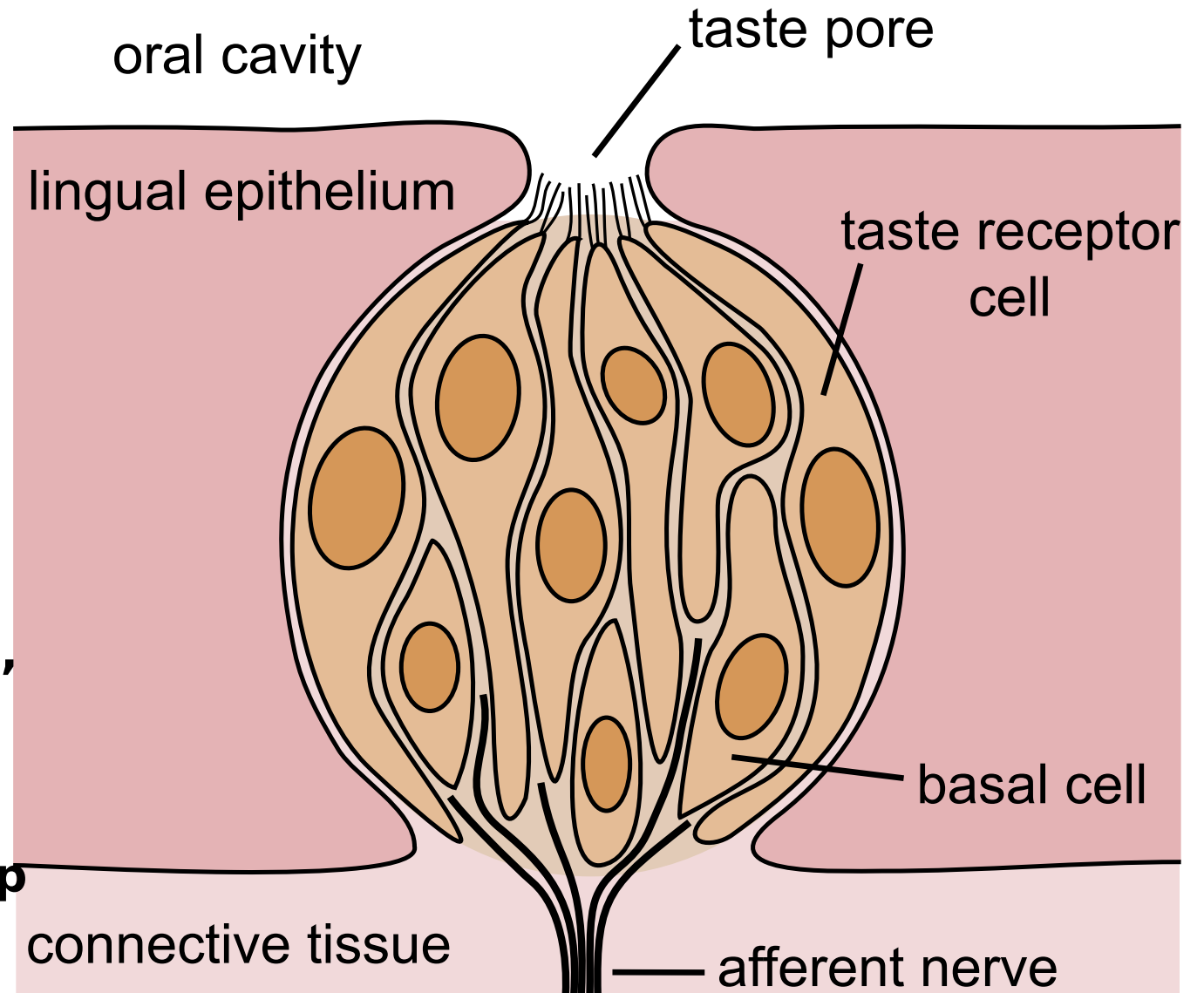
- 1- Dendrites/receptors in nasal epithelium bind to chemicals/odorants
- 2- Depolarization of **bipolar** neurons (if threshold is reached = action potential)
- 3- Bipolar neuron axons go through cribriform plate and
- 4- Synapse in the **olfactory bulb** with CN 1 (Olfactory)
- 5 - CN I innervates:
 - a- Olfactory cortex (temporal lobe)
 - b - Hypothalamus
 - c - Limbic system (emotions!)

Gustatory: Taste Buds

CHEMORECEPTOR-
phasic!

Steps:

- 1- **chemicals bind to taste bud hairs** causing **depolarization**
- 2- **three different nerves (CN 7, 9, 10)** synapse with the **medulla oblongata**
- 3 - **Then to the thalamus**
- 4- **Finally, to the insula lobe (deep in the temporal lobe)**



Know all five
basic tastes.....



REMINDERS

DUE WEDNESDAY: 1st discussion post

DUE SUNDAY:

- Pre-lesson checkpoint
- Module 6 quiz (Respondus)
- Peer response in discussion
- Lab quiz (Respondus)
- EXAM 3 opens MONDAY – covers modules 5 & 6

You can avoid feeling overwhelmed on Sunday by completing something each day – this also leads to better retention of knowledge.

Struggling?

Let's meet – send me an email and we can arrange a meeting time!

